

CI/CD for Machine Learning using MLRun

Team 06

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GITHUB REPO

1. Dataset Preparation:

- About Breast Cancer Dataset:
 - Features: 30 features describing various characteristics of cell nuclei present in breast cancer biopsies.
 - Target: Binary classification, where 0 represents benign and 1 represents malignant.
- Python Code for Data Loading:

```
from sklearn.datasets import load_breast_cancer
data = load_breast_cancer()
df = pd.DataFrame(data.data, columns=data.feature_names)
df["target"] = data.target
```

2. MLRun Project and Component Functions

- MLRun Project Initialization ([main.ipynb](#) or [Colab link](#))

A new MLRun project was initialized using the `mlrun.new_project()` method.

```
project = mlrun.new_project("lab9-breast-cancer", "./",
user_project=True, overwrite=True, init_git=False)
```

This project serves as the workspace for defining and managing all pipeline components.

- Data Preparation Script ([data_prep.py](#))

The data preparation logic was encapsulated in a separate Python script. The script loads the breast cancer dataset, processes the features and target labels, and prepares the data for model training.

```
import mlrun
from sklearn.datasets import load_breast_cancer
import pandas as pd

@mlrun.handler(outputs=["dataset", "label_column"])
def data_loader(context, format="csv"):
    """
    Loads the Breast Cancer dataset, prepares a DataFrame, and
    logs it as an MLRun Dataset artifact.
    """
    context.logger.info("Loading Breast Cancer dataset using
sklearn")
    data = load_breast_cancer()
    df = pd.DataFrame(data.data, columns=data.feature_names)
    df["target"] = data.target

    context.logger.info(f"Dataset shape: {df.shape}")
    context.logger.info(f'Saving dataset artifact using key
"breast_cancer_dataset" to {context.artifact_path}')

    # Log the DataFrame as a Dataset artifact
    context.log_dataset("breast_cancer_dataset", df=df,
format=format, index=False)

    context.logger.info("Dataset artifact logged successfully")

    return df, "target"

if __name__ == "__main__":
    with mlrun.get_or_create_ctx("data-prep-local",
upload_artifacts=True) as context:
        data_loader(context)
```

- Model Training Script ([trainer.py](#))

Model training was handled in the trainer.py script. The data was split into training and testing subsets using a 80/20 split. A Random Forest Classifier was selected as the model, and the training process was instrumented using MLRun's

apply_mlrun function to enable automatic logging and artifact tracking.

```
import mlrun
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from mlrun.frameworks.sklearn import apply_mlrun

# Define the main training function
def train_model(
    context: mlrun.run.MLClientCtx,
    dataset: mlrun.DataItem,
    label_column: str = "target",
    n_estimators: int = 100,
    max_depth: int = 5,
    random_state: int = 42,
    model_name: str = "breast_cancer_classifier"
):
    """
    Trains a RandomForestClassifier model on the input dataset.

    Uses apply_mlrun to automatically log model, metrics, and
    artifacts.
    """
    context.logger.info("Starting model training")

    # Load DataFrame from the input DataItem
    context.logger.info(f"Loading data from {dataset.url}")
    df = dataset.as_df()
    context.logger.info(f>Data loaded, shape: {df.shape}")

    # Prepare features (X) and target (y)
    X = df.drop(label_column, axis=1)
    y = df[label_column]
    context.logger.info(f"Features shape: {X.shape}, Target
    shape: {y.shape}")

    # Split data into training and testing sets
    X_train, X_test, y_train, y_test = train_test_split(X, y,
    test_size=0.2, random_state=random_state)
    context.logger.info(f"Train set size: {X_train.shape[0]},
    Test set size: {X_test.shape[0]}")

    # Initialize the RandomForestClassifier model
    model = RandomForestClassifier(
        n_estimators=n_estimators,
        max_depth=max_depth,
        random_state=random_state
    )
```

```

        context.logger.info(f"Initialized RandomForestClassifier
with n_estimators={n_estimators}, max_depth={max_depth}")

        # Log the model and metrics using apply_mlrun
        apply_mlrun(model=model, model_name=model_name,
x_test=X_test, y_test=y_test)
        model.fit(X_train, y_train)
        context.logger.info("Training completed and model artifacts
logged via apply_mlrun.")

```

- Model Serving Script ([serving.py](#))

Model serving was implemented by defining a class that extends `mlrun.serving.V2ModelServer`. The class includes the `load()` and `predict()` methods, allowing seamless model deployment and prediction serving.

```

import mlrun
from cloudpickle import load
import numpy as np
from typing import List

class BreastCancerModel(mlrun.serving.V2ModelServer):
    """
    MLRun Serving Class for the Breast Cancer RandomForest
    model.
    Inherits from V2ModelServer for standard model serving
    capabilities.
    """

    def load(self):
        """
        Loads the trained RandomForest model from the specified
        path using get_model().
        """
        self.context.logger.info(f>Loading model from
{self.model_path}...)
        model_file, extra_data = self.get_model('.pkl')
        self.model = load(open(model_file, 'rb'))
        self.context.logger.info("Model loaded successfully.")

    def predict(self, body: dict) -> List:
        """
        Receives prediction requests and returns model
        predictions.
        Assumes that the 'inputs' key in the body contains a
        list of feature lists.
        """

```

```
feats = np.asarray(body['inputs'])
self.context.logger.info(f"Received {feats.shape[0]}
samples for prediction.")
results: np.ndarray = self.model.predict(feats)
return results.tolist()
```

- Workflow Definition ([workflow.py](#))

The end-to-end ML pipeline was constructed using the `@dsl.pipeline` decorator. This workflow orchestrates three stages: data ingestion, model training with hyperparameter tuning, and model deployment.

Hyperparameter tuning was conducted over `n_estimators: [10, 100, 200]` and `max_depth: [2, 5, 10]`, selecting the model with the highest accuracy.

Model deployment was achieved using `mlrun.deploy_function`, targeting the base Kubernetes image `mlrun/mlrun`.

```
import mlrun
from kfp import dsl

@dsl.pipeline(name="breast-cancer-demo")
def pipeline(model_name="breast-cancer-classifier"):

    ingest = mlrun.run_function(
        "load-breast-cancer-data",
        name="load-breast-cancer-data",
        params={"format": "pq", "model_name": model_name},
        outputs=["dataset"],
    )

    train = mlrun.run_function(
        "trainer",
        inputs={"dataset": ingest.outputs["dataset"]},
        hyperparams={
            "n_estimators": [10, 100, 200],
            "max_depth": [2, 5, 10]
        },
        selector="max.accuracy",
        outputs=["model"],
    )

    deploy = mlrun.deploy_function(
        "serving",
        models=[{"key": model_name, "model_path":
train.outputs["model"], "class_name": "BreastCancerModel"}],
```

```
mock=True  
)
```

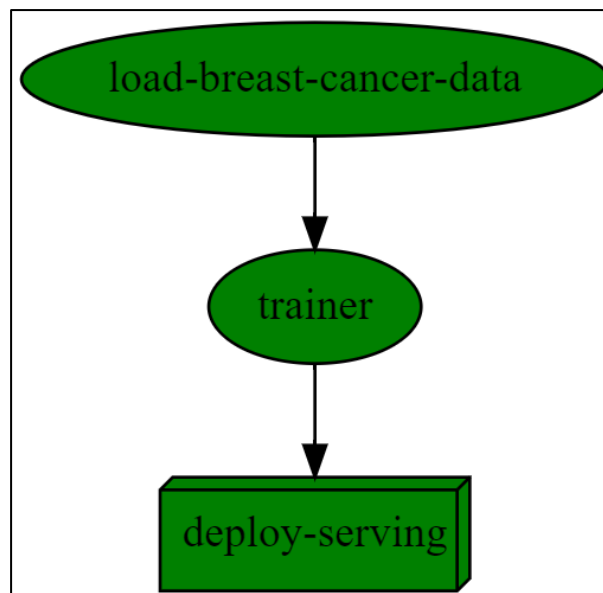
3. Workflow Execution and Artifacts

- Workflow Graph

Upon execution using

```
ru_id = project.run(  
    workflow_path="./workflow.py",  
    arguments={"model_name": "breast-cancer-classifier"},  
    watch=True, local=no_k8s)
```

, a visual representation of the pipeline was generated, illustrating the connections between data ingestion, model training, and deployment stages.



- Data Preparation Artifact

The dataset was loaded using the `load-breast-cancer-data` function, which outputs the dataset as an artifact in MLRun. The artifact was retrieved from the `gen_data_run` object and

displayed using `gen_data_run.artifact("dataset").as_df().head()`.

```
gen_data_run = project.run_function("load-breast-cancer-data", params={"format":"csv"}, local=True)
```

✓ 36.8s Python

```
> 2025-04-14 15:08:42,342 [warning] it is recommended to use k8s secret (specify secret_name), specifying the aws_access_key/aws_secret_key
> 2025-04-14 15:08:42,427 [info] Storing function: {"db":"http://localhost:30070","name":"load-breast-cancer-data","uid":"caaed6bb93a946f68a"}
> 2025-04-14 15:08:43,050 [info] Handler was not provided running main (data_prep.py)
Running: ['c:\\Users\\adibo\\anaconda3\\envs\\sttai\\python.exe', '-u', 'data_prep.py']
> 2025-04-14 15:09:04,735 [info] logging run results to: http://localhost:30070
> 2025-04-14 15:09:04,871 [info] Loading Breast Cancer dataset using sklearn
> 2025-04-14 15:09:04,913 [info] Dataset shape: (569, 31)
> 2025-04-14 15:09:04,914 [info] Saving dataset artifact using key "breast_cancer_dataset" to s3://mlrun/projects/lab9-breast-cancer-adibo/
> 2025-04-14 15:09:13,881 [info] Dataset artifact logged successfully
```

project	uid	iter	start	state	kind	name	labels	inputs	parameters	results	artifacts
lab9-breast-cancer-adibo	...f3b08f	0	Apr 14 09:39:04	completed	run	load-breast-cancer-data	kind=local owner=adibo host=Adi		format=csv label_column=target		breast_cancer_dataset

```
gen_data_run.artifact("dataset").as_df().head()
```

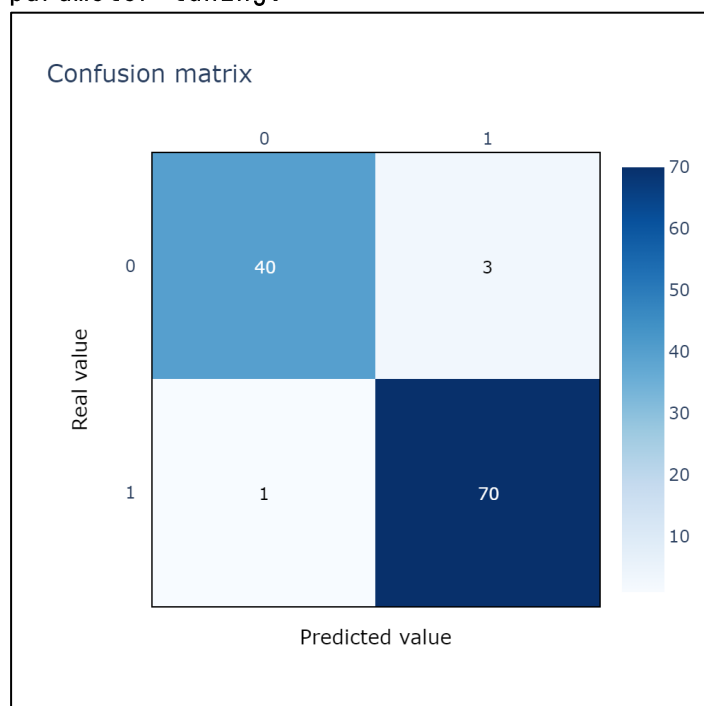
✓ 2.8s Python

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	worst texture	worst perimeter	worst area	worst smoothness
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710	0.2419	0.07871	...	17.33	184.60	2019.0	0.1
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667	...	23.41	158.80	1956.0	0.1
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790	0.2069	0.05999	...	25.53	152.50	1709.0	0.1
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	0.2597	0.09744	...	26.50	98.87	567.7	0.1
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430	0.1809	0.05883	...	16.67	152.20	1575.0	0.1

5 rows x 31 columns

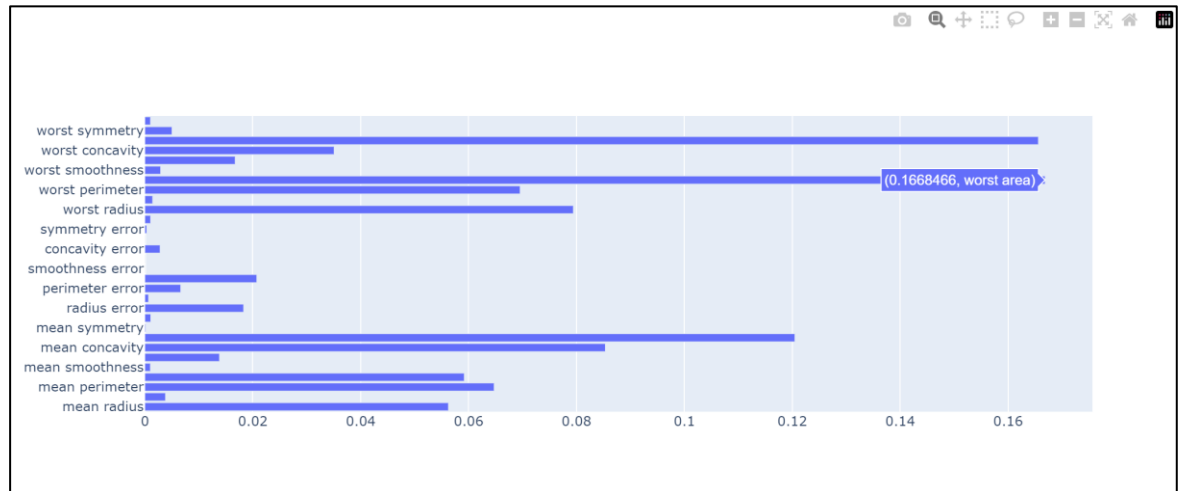
- Training Metrics - Confusion Matrix

During model training, MLRun automatically generated performance artifacts, including the confusion matrix. This is the confusion matrix of the best model found after hyper-parameter tuning.



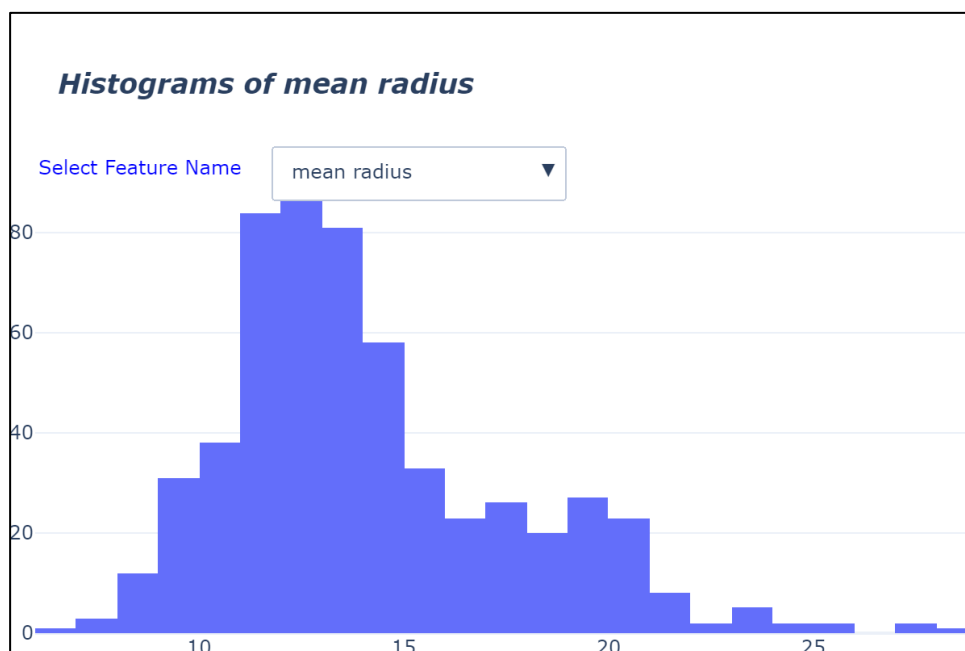
- Feature Selection Artifact

MLRun logged feature importances via Random Forest's built-in mechanism. This artifact suggests that 'worst area' is the feature of highest importance.

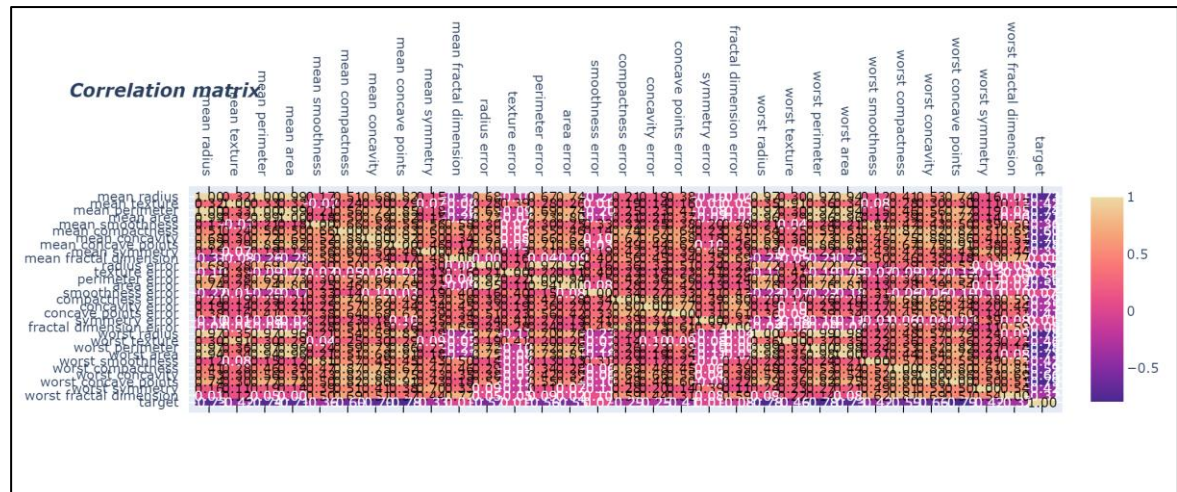


#Extras

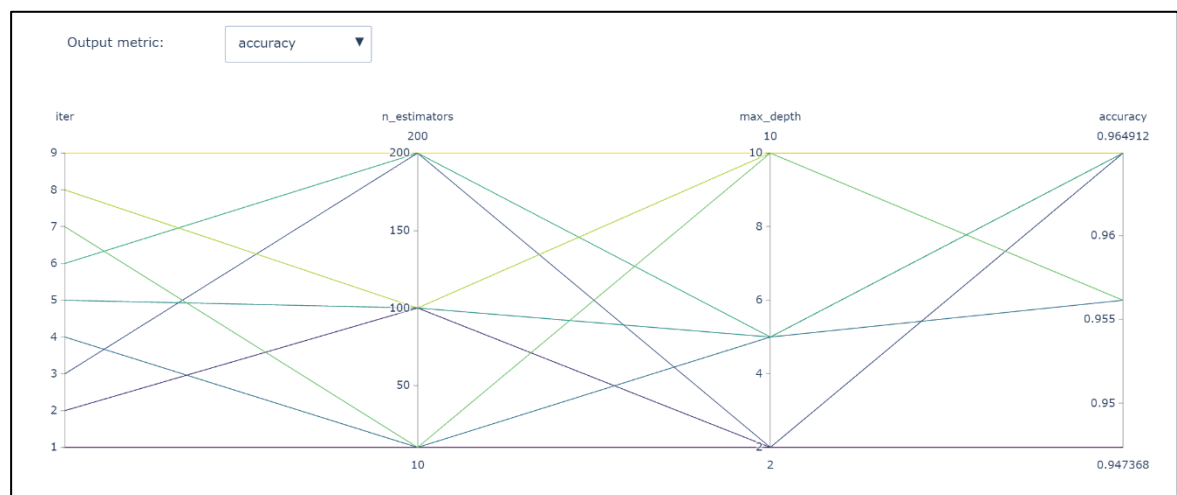
- Histograms of features of the dataset



- Correlation matrix of the dataset



- Parallel Coordinates Artifact



This suggests keeping accuracy as the metric, the best set of hyperparameters are:

Iteration 2: n_estimators: 100, max_depth = 2
 Iteration 3: n_estimators: 200, max_depth = 2
 Iteration 5: n_estimators: 100, max_depth = 5
 Iteration 6: n_estimators: 200, max_depth = 5
 Iteration 8: n_estimators: 100, max_depth = 10
 Iteration 9: n_estimators: 200, max_depth = 10

All performing equally well with an accuracy of 0.9649122807017544 on test dataset.

- MLRun UI:

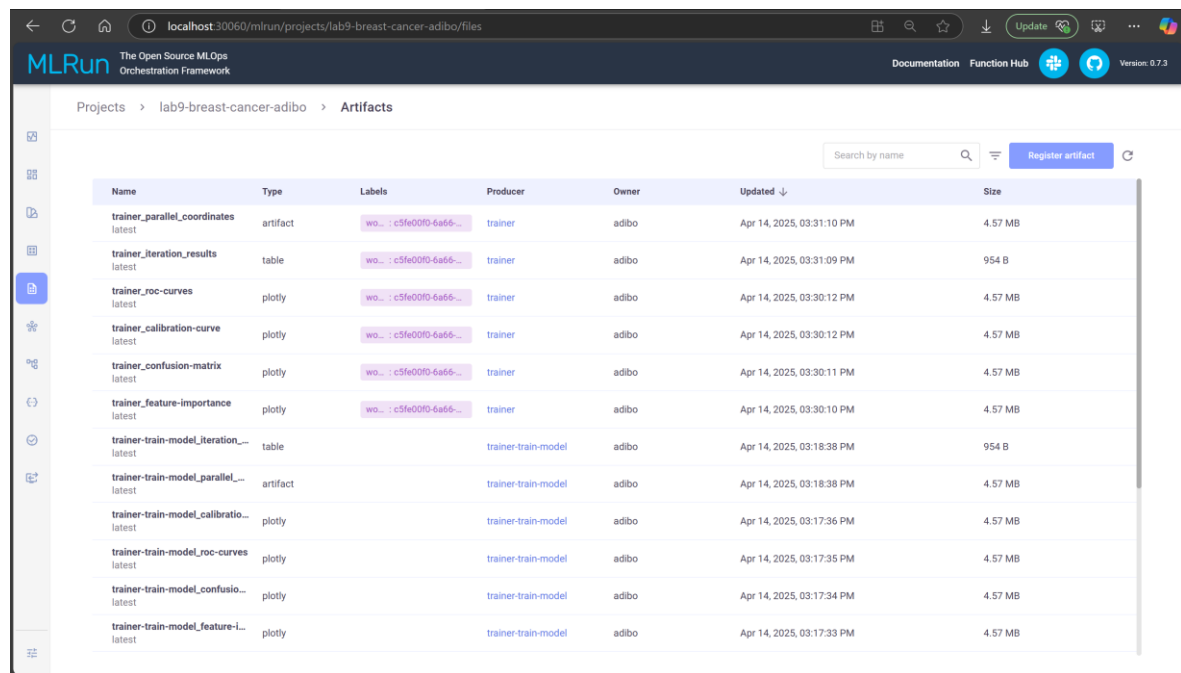
The top screenshot shows the MLRun UI for the project 'lab9-breast-cancer-adibo'. The page displays various counters: Models (1), Feature sets (0), Artifacts (19), Running jobs (0), Running workflows (0), Failed (0), and Scheduled (0). It also shows a 'Recent' table with columns: Name, Type, Status, Started at, and Duration. The table lists jobs like 'trainer', 'load-breast-cancer-data', 'trainer-train-model', and 'task-describe'.

The bottom screenshot shows the 'load-breast-cancer-data' job details page. It includes tabs for Overview, Inputs, Artifacts, Results, Logs, and Pods. The 'Artifacts' tab is active, showing a table with columns: Name, Path, Size, and Updated. The table lists artifacts like 'load-breast-cancer-data_breast_c...'. Below this, there is a data table with columns: index, mean radius, mean texture, mean perimeter, mean area, mean smooth..., mean compac..., mean concavity, mean concav..., and mea.

Name	Type	Status	Started at	Duration
trainer			Apr 14, 2025, 03:29:57 PM	00:01:12
load-breast-cancer-data			Apr 14, 2025, 03:29:16 PM	00:00:06
trainer-train-model			Apr 14, 2025, 03:17:19 PM	00:01:19
task-describe			Apr 14, 2025, 03:11:32 PM	00:00:16

Name	Path	Size	Updated
load-breast-cancer-data_breast_c...	s3://mirun/projects/lab9-breast-cancer-adibo/artifacts/c5fe00f0-6a...	121 kB	Apr 14, 2025, 03:29:1...

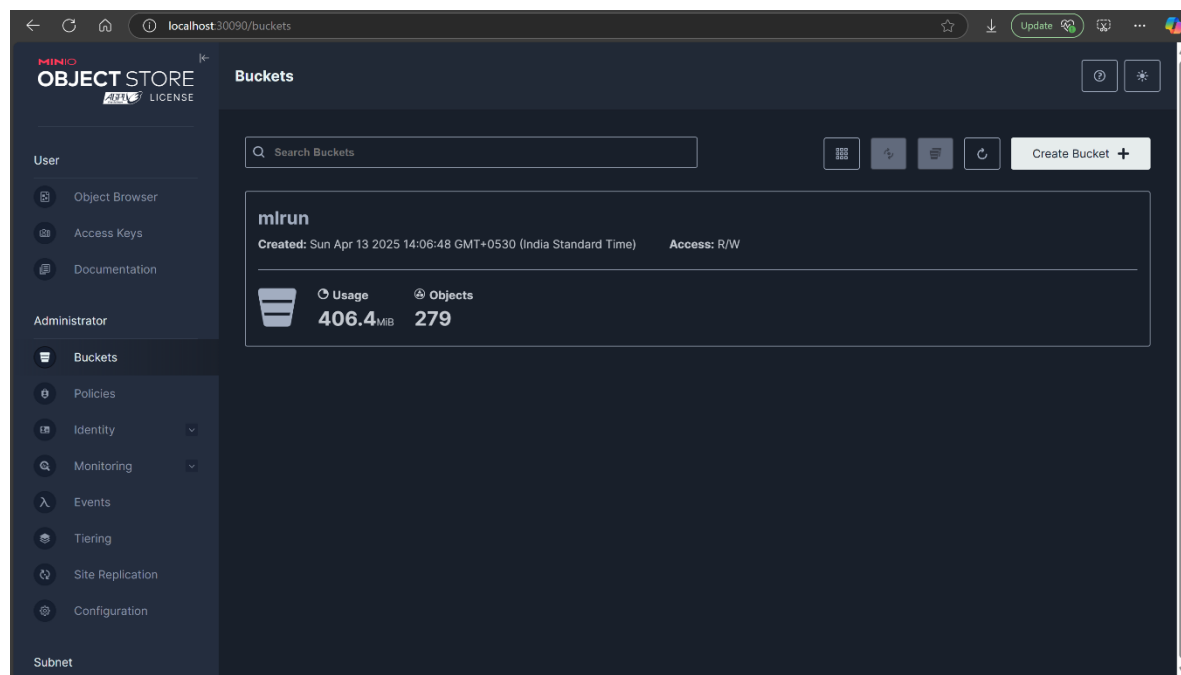
index	mean radius	mean texture	mean perimeter	mean area	mean smooth...	mean compac...	mean concavity	mean concav...	mea
0	17.99	10.38	122.8	1001	0.1184	0.2776	0.3001	0.1471	0.24
1	20.57	17.77	132.9	1326	0.08474	0.07864	0.0869	0.07017	0.18
2	19.69	21.25	130	1203	0.1096	0.1599	0.1974	0.1279	0.20



The screenshot shows the MLRun web interface at localhost:30060. The 'Artifacts' page displays a table of training artifacts for the 'lab9-breast-cancer-adibo' project. The table includes columns for Name, Type, Labels, Producer, Owner, Updated, and Size. The artifacts are listed in descending order of update time.

Name	Type	Labels	Producer	Owner	Updated	Size
trainer_parallel_coordinates latest	artifact	wo... : c5fe00f0-6a65-...	trainer	adibo	Apr 14, 2025, 03:31:10 PM	4.57 MB
trainer_iteration_results latest	table	wo... : c5fe00f0-6a65-...	trainer	adibo	Apr 14, 2025, 03:31:09 PM	954 B
trainer_roc-curves latest	plotly	wo... : c5fe00f0-6a65-...	trainer	adibo	Apr 14, 2025, 03:30:12 PM	4.57 MB
trainer_calibration-curve latest	plotly	wo... : c5fe00f0-6a65-...	trainer	adibo	Apr 14, 2025, 03:30:12 PM	4.57 MB
trainer_confusion-matrix latest	plotly	wo... : c5fe00f0-6a65-...	trainer	adibo	Apr 14, 2025, 03:30:11 PM	4.57 MB
trainer_feature-importance latest	plotly	wo... : c5fe00f0-6a65-...	trainer	adibo	Apr 14, 2025, 03:30:10 PM	4.57 MB
trainer-train-model_iteration_... latest	table		trainer-train-model	adibo	Apr 14, 2025, 03:18:38 PM	954 B
trainer-train-model_parallel_... latest	artifact		trainer-train-model	adibo	Apr 14, 2025, 03:18:38 PM	4.57 MB
trainer-train-model_calibratio... latest	plotly		trainer-train-model	adibo	Apr 14, 2025, 03:17:36 PM	4.57 MB
trainer-train-model_roc-curves latest	plotly		trainer-train-model	adibo	Apr 14, 2025, 03:17:35 PM	4.57 MB
trainer-train-model_confusio... latest	plotly		trainer-train-model	adibo	Apr 14, 2025, 03:17:34 PM	4.57 MB
trainer-train-model_feature-l... latest	plotly		trainer-train-model	adibo	Apr 14, 2025, 03:17:33 PM	4.57 MB

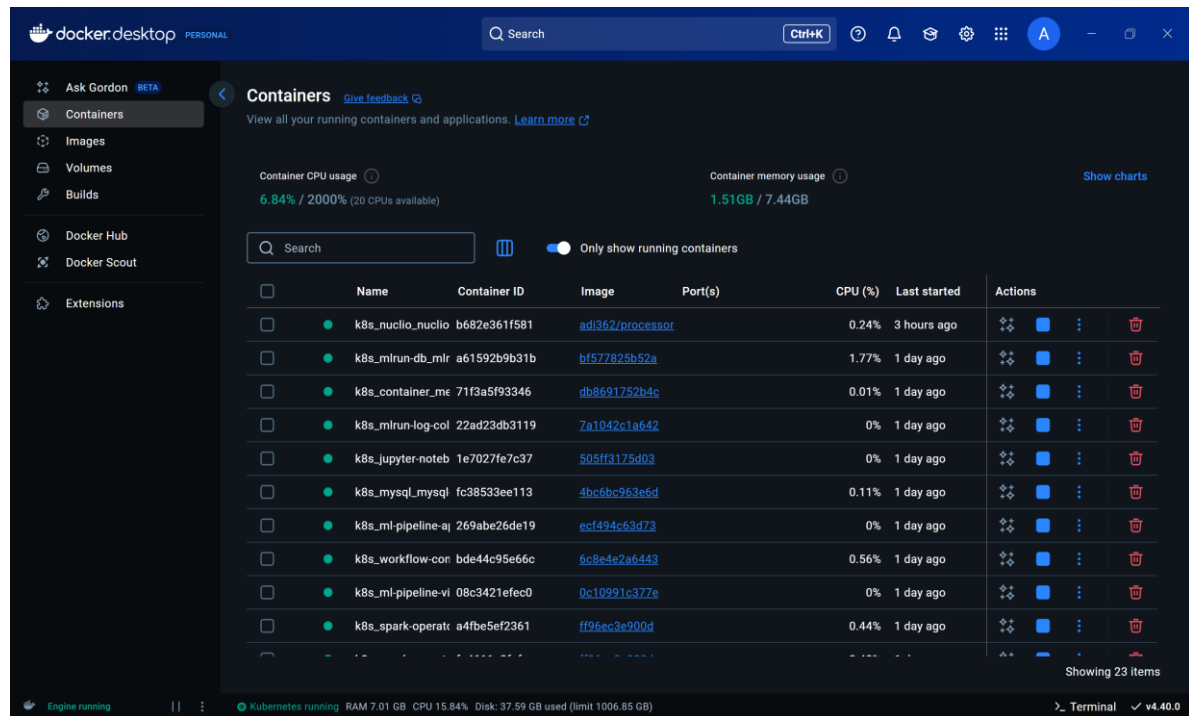
- Minio UI



The screenshot shows the Minio Object Store web interface at localhost:30090. The 'Buckets' page displays a list of buckets. The 'mlrun' bucket is highlighted, showing its creation time and access permissions. The bucket's usage and object count are also displayed.

Bucket	Created	Access	Usage	Objects
mlrun	Sun Apr 13 2025 14:06:48 GMT+0530 (India Standard Time)	R/W	406.4 MiB	279

- Docker Desktop



- Verification

```
(sttai) C:\Users\adibo>kubectl get pods -n mlrun
NAME                                READY   STATUS    RESTARTS   AGE
breast-cancer-demo-gw72m-2034386729 0/2     Completed 0           3h36m
breast-cancer-demo-gw72m-3764670050 0/2     Completed 0           3h34m
breast-cancer-demo-gw72m-3863584672 0/2     Completed 0           3h37m
load-breast-cancer-data-f665f        0/1     Completed 0           3h37m
metadata-envoy-deployment-86b9c85656-48b4w 1/1     Running   0           28h
metadata-grpc-deployment-5dfd66bb8d-r9ck5 1/1     Running   1 (28h ago) 28h
metadata-writer-7b7fc9f477-zgs4x      1/1     Running   0           28h
minio-65c9b8dc44-bbwcw               1/1     Running   0           28h
ml-pipeline-5d7998ffc5-pt5zf          1/1     Running   0           28h
ml-pipeline-persistenceagent-69446f4bdd-jp55d 1/1     Running   0           28h
ml-pipeline-scheduledworkflow-8564b98f5b-nstrs 1/1     Running   0           28h
ml-pipeline-ui-5d9ff6dc69-lc68m       1/1     Running   0           28h
ml-pipeline-viewer-crd-6668f68cd7-wcphk 1/1     Running   0           28h
ml-pipeline-visualizationserver-6878c5b646-bnz6m 1/1     Running   0           28h
mlrun-api-chief-86455dc6d7-q4qjj      2/2     Running   0           28h
mlrun-db-6c85956db5-vx9jg            1/1     Running   0           28h
mlrun-jupyter-85768bb958-hzf5g       1/1     Running   0           28h
mlrun-ui-744bcb7f44-tvd5g            1/1     Running   0           28h
mpi-operator-97dd956c5-qxwpn          1/1     Running   0           28h
mysql-9dc78bdd7-f6ttg                1/1     Running   0           28h
nuclio-controller-94fff55d7-hvfmb     1/1     Running   0           28h
nuclio-dashboard-6ddd759444-d8t98     1/1     Running   0           28h
nuclio-lab9-breast-cancer-adibo-serving-6f796dff54-xdbz4 1/1     Running   0           3h25m
spark-operator-controller-855dc75b69-mlpkm 1/1     Running   0           28h
spark-operator-webhook-587b54f56-ws9tq 1/1     Running   0           28h
task-describe-cjxst                  0/1     Completed 0           3h54m
trainer-j4pvv                         0/1     Completed 0           3h36m
trainer-train-model-f9swp             0/1     Completed 0           3h48m
workflow-controller-789b47d74-7kw6g   1/1     Running   0           28h
```

```
(sttai) C:\Users\adibo>kubectl get deployments -n mlrun
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
metadata-envoy-deployment	1/1	1	1	28h
metadata-grpc-deployment	1/1	1	1	28h
metadata-writer	1/1	1	1	28h
minio	1/1	1	1	28h
ml-pipeline	1/1	1	1	28h
ml-pipeline-persistenceagent	1/1	1	1	28h
ml-pipeline-scheduledworkflow	1/1	1	1	28h
ml-pipeline-ui	1/1	1	1	28h
ml-pipeline-viewer-crd	1/1	1	1	28h
ml-pipeline-visualizationserver	1/1	1	1	28h
mlrun-api-chief	1/1	1	1	28h
mlrun-api-worker	0/0	0	0	28h
mlrun-db	1/1	1	1	28h
mlrun-jupyter	1/1	1	1	28h
mlrun-ui	1/1	1	1	28h
mpi-operator	1/1	1	1	28h
mysql	1/1	1	1	28h
nuclio-controller	1/1	1	1	28h
nuclio-dashboard	1/1	1	1	28h
nuclio-lab9-breast-cancer-adibo-serving	1/1	1	1	3h38m
spark-operator-controller	1/1	1	1	28h
spark-operator-webhook	1/1	1	1	28h
workflow-controller	1/1	1	1	28h

```
(sttai) C:\Users\adibo>kubectl get services -n mlrun
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
metadata-envoy-service	ClusterIP	10.110.57.166	<none>	9090/TCP	28h
metadata-grpc-service	ClusterIP	10.106.1.175	<none>	8080/TCP	28h
minio	NodePort	10.96.204.79	<none>	9000:30080/TCP	28h
minio-console	NodePort	10.100.115.235	<none>	9001:30090/TCP	28h
ml-pipeline	ClusterIP	10.101.9.138	<none>	8888/TCP,8887/TCP	28h
ml-pipeline-ui	NodePort	10.98.186.219	<none>	80:30100/TCP	28h
ml-pipeline-visualizationserver	ClusterIP	10.106.68.119	<none>	8888/TCP	28h
mlrun-api	NodePort	10.106.97.115	<none>	8080:30070/TCP	28h
mlrun-api-chief	ClusterIP	10.111.2.34	<none>	8080/TCP	28h
mlrun-db	ClusterIP	10.105.220.134	<none>	3306/TCP	28h
mlrun-jupyter	NodePort	10.101.16.112	<none>	8888:30040/TCP	28h
mlrun-ui	NodePort	10.98.3.206	<none>	80:30060/TCP	28h
mysql	ClusterIP	10.98.30.23	<none>	3306/TCP	28h
nuclio-dashboard	NodePort	10.98.104.3	<none>	8070:30050/TCP	28h
nuclio-lab9-breast-cancer-adibo-serving	NodePort	10.103.25.249	<none>	8080:30340/TCP	3h38m
spark-operator-webhook-svc	ClusterIP	10.109.217.218	<none>	9443/TCP	28h
workflow-controller-metrics	ClusterIP	10.107.186.158	<none>	9090/TCP	28h
